Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/17/2018
PI Name:	Kassemi, Mohammad Ph.D.		
Project Title:	The Zero Boil-Off Tank (ZBOT) Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE (Dec 2019): former affiliation included National Center for Space Exploration Research (NCSER), per information from J. McQuillen/GRC		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/31/2018 per J. McQuillen/GRC (Ed., 5/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chato, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	Internal Project		
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Task Description:	Principal Investigator (PI): Dr. Mohammad Kassemi (Case Western Reserve University, NCSER/NASA Glenn Research Center (GRC)); Co-Investigator: Dr. David Chato (NASA GRC); Project Scientist: John Mcquillen (NASA GRC); Project Manager: Bill Sheredy (NASA GRC). NASA's projected exploration program includes a series of human and robotic expeditions to low and high Earth orbit, Moon, Mars, and possibly the asteroids and moons of other planets. Integral to all phases of these space and planetary expeditions is affordable and reliable cryogenic fluid storage for use in the propellant or life support systems. Without safe, efficient, and flexible cryogen storage, economically justified human missions may not be possible. Cryogen vaporization caused by various heat leaks into the tank from its surroundings is the main cause of storage tank self-pressurization. This has led to the development of innovative pressure control designs based on mixing of the bulk liquid with or without active cooling. Unfortunately, both tank pressurization and pressure control are governed by intricate and complicated dynamic interactions among the forced mixing, the various gravity-dependent or -independent transport mechanisms in the vapor and liquid phases, and the condensation/evaporation process at the interface. Consequently, effective implementation and optimization of a dynamic pressure control system for space applications cannot be accomplished solely by empiricism, especially, in view of serious scarcity of relevant microgravity data. Since large-scale in-space technology validation test or demonstration of the cryogenic storage tank prototype has not been possible due to both budgetary and time constraints, it was proposed that a series of small-scale targeted microgravity simulant fluid storage tank experiments can be performed aboard the International Space Station (ISS) to first understand the underlying scientific physical phenomena influencing tank pressurization/pressure control in space and then to validate and verify state-of-the-art two-phase computational fluid dynamics (CFD) models that can be used to scale-up the storage tank pressure control design for various in-orbit or on-surface applications. In this manner, the risks associated with the design of space-based storage tanks using only ground testing of the technology can be greatly reduced. In this light, the Zero-Boil-Off Tank (ZBOT) experiments have been developed as a series of small scale ISS experiments aimed at delineating important fluid flow, heat and mass transport, and phase change phenomena that affect cryogenic storage tank pressurization and pressure control in microgravity. The experiments use a simulant transparent low boiling point fluid (PnP) in a sealed transparent Dewar to study and quantify: (a) fluid flow and thermal stratification during pressurization; (b) mixing, thermal destratification, depressurization, and jet-ullage penetration during pressure control; (c) transport and kinetics effects of noncondensable pressurant gases on interfacial evaporation and condensation rates; and (d) transport, heat transfer, and phase change mass transfer interactions between evaporating and condensing sprayed droplets and ullage during active spray-bar-droplet cooling. The experiments will provide valuable microgravity empirical two-phase data associated with the above-mentioned physical phenomena through highly accurate local wall and fluid temperature and pressure measurements, nonintrusive Particle Imaging Velocimetry (PIV), and full-field phase-distribution and flow visualization. Moreover, the experiments are performed under tightly controlled and definable heat transfer boundary conditions to provide reliable high-fidelity data and precise input as required for validation & verification of state-of-the-art two-phase CFD models developed as part of this research and by other groups in the international scientific and cryogenic fluid management communities. During the project review process, the ZBOT Science Review Panel consisting of members from academia, NASA space & flight centers, and aerospace industry recommended that the broad scientific goals of the project can best be achieved through a series of consecutive experiments. Thus, the first experiment (ZBOT-1) in the series is focused on microgravity tank pressurization, thermal stratification, and destratification through mixing. The second experiment (ZBOT-2) aims at delineating the important effects of noncondensable pressurants on tank pressurization and pressure control. And the third experiment (ZBOT-3) investigates transport and phase change aspects of active tank pressure control by spray-droplet and forced-jet mixing/cooling. Implementation of the recommended strategy has resulted in a logical hierarchical series of three consecutive experiments whereupon the main tank, the fluid support unit, and the diagnostic rig are constructed during ZBOT-1 and augmented in a cost-effective step-by-step manner through implementation of additional units and diagnostics to accommodate the added complexities required by ZBOT-2 and ZBOT-3 experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The result of the ZBOT investigation will primarily and directly address the fundamental need of NASA for efficient, reliable, and enabling propellant and life support technologies for long-duration planetary exploration. As such, ZBOT1-3 provide specific fundamental scientific knowledge and understanding that is needed to improve key cryogenic fluids management technologies for storage and transfer of propellants during Lunar Gateway and evolvable Mars Missions. However, the fundamental two-phase data and scientific understanding provided by the experiment and the associated theoretical model development will also affect a diverse set of ground-based two-phase technologies such as condensing heat exchangers, phase separators, water-handling operations in fuel cells, and storage of cryogenic and non cryogenic consumable fluids.
	ZBOT 2017-2019 Task Progress The Zero-Boil-Off Tank (ZBOT) Experiments are a series of small scale science experiments aboard the International Space Station (ISS) that use a transparent volatile simulant fluid in a transparent sealed tank to delineate various fluid flow, heat and mass transport, and phase change phenomena that control storage tank pressurization and pressure control in microgravity. The hardware flew to ISS on the OA-7 flight on April 18th 2017 and operations were initiated in September 2017. The experiment lasted roughly 3 months and encompassed more than 90 tests that studied fluid flow and thermal stratification during the self-pressurization interval, and thermal de-stratification, depressurization, mixing, and jet ullage penetration dynamics during the pressure control interval. Hand-in-Hand with the experiments, comprehensive two-phase Computational Fluid Dynamics (CFD) models were also developed for simulation and prediction of storage tank pressurization and pressure control. The models use either a sharp interface or a volume of fluid (VOF) interface capturing scheme to predict the evolution of fluid velocities, temperatures, and multi-component species concentrations in the liquid and vapor phases under tight coupling with the evaporative/condensing mass transfer and energy and force balances at the phase front. At this stage, the CFD models are partially verified and validated with ZBOT ground-based and microgravity data and existing data from large-scale ground-based LH2 experiments performed by NASA. The models are currently being further validated by the extensive

Task Progress:	1g and microgravity data provided by the ZBOT-1 experiments. Microgravity Tests Performed: Over 90 microgravity tests were performed at approximately 70%, 80%, and 90% tank fill levels in a span of roughly 3 month from October – December 2017. Self-pressurization tests were conducted under three modes: VJ (vacuum jacket) heating, strip heating and simultaneous VJ and strip heating in an attempt to simulate heat leaks from the environment, the support structure and both. In order to have consistent data conducive for model validation, all the self-pressurization tests were performed from the same well-established and quantifiable uniform temperature initial condition established by intervals of constant temperature mixing and hold during the pre-test tank preparation. The pressure control studies were performed either from an elevated uniform temperature condition or from thermally stratified conditions following a self-pressurization run. Liquid is drawn from the tank, passed through the heat exchanger in the Fluid Support Unit (FSU), and then injected into the tank at a given flow rate either at the average tank fluid temperature for isothermal mixing and ullage penetration studies or at a prescribed subcooled level for active cooling pressure control tests. Jet flow rates were varied from 4-25 cm/s spanning a range of jet Reynolds number in laminar, transitional, and turbulent regimes and a range of Weber numbers covering no ullage penetration, partial penetration and complete ullage penetration and break-up. Microgravity jet mixing and ullage penetration CFD simulations were performed by our two-phase 3D VOF model and used to guide the design of the microgravity experimental test matrix. The digital imaging and machine readable textual/numerical data from the experiments were continuously downloaded. Both the raw experimental data and a set of reduced data analyzed and processed by the ZBOT science team together with the CFD simulations results will be stored on the ISS Physical Sciences Open System Repository Server operated and maintained by NASA. Approximately one year after the end of ZBOT's microgravity operations, this data will be released for use to the entire scientific and engineering communities at large. Summary of Important Microgravity Test Results: The experimental data was processed, analyzed, plotted, and interpreted carefully. Numerical CFD predictions were also performed and compared to the experimental results to validate the two-phase storage tank CFD model and to gain insight into the unexpected and non-intuitive microgravity behavior, especially, during the jet mixing operations. From the experimental and numerical results, examined to date, the following conclusions can be drawn: 1. For a sealed tank designed and operating under low heat loads, self-pressurization in microgravity proceeds under near thermodynamic equilibrium conditions and can be captured by the two-phase CFD models with good fidelity using either thermo-based or kinetic–based interfacial mass transfer submodels. 2. Microgravity self-pressurization essentially follows the same behavior and trend as 1g but at lower rates and magnitudes. Thus ground-based testing can provide a useful conservative estimate for flight pressurization rates. 3. During jet mixing, the behavior and interaction of the jet with the ullage in microgravity is non-intuitive and quite different at any jet velocity from its 1g counterpart. 4. At high jet flow rate mixing, the ullage position is maintained at the top of the enclosure. The ullage will deform and maneuver to accommodate the jet passage with only partial penetration. The envisioned full penetration and splitting of the ullage may not occur easily during space operations due to lack of perfect jet bubble alignment. 5. At low flow rate mixing, and with less than perfect jet alignment, the ullage will follow a non-intuitive downward motion (against both residual gravity and jet flow direction) towards the jet nozzle inlet to the tank. This points to the possibility of using a forced jet flow with a minor tilt to move and relocate the ullage and thus control its position in the tank during microgravity operations. Archival Publication & Conference Presentations: In 2017-2018 the Principal Investigator (PI) and Science Team presented the results of ZBOT ground-based activity in 2 archival journal publications and 10 conference presentation/papers. The following post-flight NASA & industry Webex/Presentations were also undertaken: • Webex Presentation to Southwest Research Institute, Dec 14 2017 ; • Webex Presentation to NASA AES (Advanced Exploration Systems) Lunar Lander Management (Suzuki Nantel and Greg Chavers) on April 5th 2018 ; • Presentation to NASA NESC Cryogenics Technical Discipline Team Meeting, Kennedy Space Center, April 26th 2018 ; • Webex Presentation to Blue Origin, May 3rd 2018 ; • Presentation to NASA SLPSRA (Space Life & Physical Sciences Research & Applications) Division, June 14th 2018 ; • Webex Presentation to NASA STMD (Science Technology Mission Directorate) Management, NASA Glenn Research Center (GRC), June 15th 2018 ; • Webex Presentation/LAD Discussion with Orbital-ATK, PMD Technologies, June 22nd 2018 • Two Recorded Webcast Presentations sponsored by NASA Engineering and Safety Center (NESC) Academy, June 2018 & TBD ;The first NESC Webcast presented in June 2018 had around 150 national and international attendees. Linkage with NASA Space Technology Mission Directorate (STMD) & NASA Exploration Program: In 2017-2018, ZBOT microgravity data played a large role in validation of the CFD models and codes developed under the Development and Validation of Analysis Tools (DVAT) task of the NASA STMD's Evolvable Cryogenics (eCryo) project. In addition to the CFD model validations undertaken by the ZBOT PI and science team two different classes of models are also being validated under the DVAT task of the eCryo project. The first is another CFD model developed using the framework of the FLOW3D industry standard multipurpose code. The second consists of multi-node models developed using system level ThermalDesktop and Sinda/Fluint engineering codes.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Abstracts for Journals and Proceedings	Kassemi M, Hylton S, Kartuzova O. "Zero Boil-Off Tank (ZBOT) Experiment & CFD Model Validation." The 65th JANNAF Propulsion Meeting, Long Beach, Ca, May 21-24 2018. 65th JANNAF Propulsion Meeting, Long Beach, Ca, May 21-24 2018. , May-2018

Abstracts for Journals and Proceedings	Kassemi M, Hylton S, Kartuzova O. "Zero-Boil-Off Tank (ZBOT) Experiment – Ground-Based Validation of Self-Pressurization & Pressure Control Two-Phase CFD Model." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kassemi M, Kartuzova O, Hylton S. "Validation of Storage Tank Self-Pressurization & Pressure Control CFD Models: Important Considerations Revealed in Crossing Fluid Types, Scales, and Gravity Levels." The 27th Space Cryogenics Workshop, Oak Brook, Illinois, July 5-7, 2017. Proceedings of the 27th Space Cryogenics Workshop, Oak Brook, Illinois, July 5-7, 2017. , Jul-2017
Articles in Peer-reviewed Journals	Kassemi M, Kartuzova O, Hylton S. "Validation of two-phase CFD models for propellant tank self-pressurization: Crossing fluid types, scales, and gravity levels." Cryogenics. 2018 Jan;89:1–15. https://doi.org/10.1016/j.cryogenics.2017.10.019 , Jan-2018
Papers from Meeting Proceedings	Kassemi M, Hylton S, Kartuzova O. "Zero-Boil-Off Tank Experiment - 1G and Microgravity Pressurization & Pressure Control Results." 69th International Astronautical Congress (IAC), Bremen, Germany, October 1-5, 2018. 69th International Astronautical Congress (IAC), Bremen, Germany, October 1-5, 2018. , Oct-2018
Papers from Meeting Proceedings	Kartuzova O, Kassemi M. "A Computational Study of the Effect of Sloshing with High Level of Lateral Acceleration on the Heat Transfer and Pressure Drop in a Small-Scale Tank in Normal Gravity." ASME 5th Joint US-European Fluids Engineering Summer Conference (FEDSM2018), Montreal, Canada, July 15-20, 2018. ASME 5th Joint US-European Fluids Engineering Summer Conference (FEDSM2018), Montreal, Canada, July 15-20, 2018. , Jul-2018
Papers from Meeting Proceedings	Kassemi M, Kartuzova O, Hylton S. "Zero-Boil-Off Tank (ZBOT) Experiment – Ground-Based Validation of Two – Phase Self-Pressurization CFD Model & Preliminary Microgravity Results." 2018 Joint Propulsion Conference, AIAA Propulsion and Energy Forum, Cincinnati, OH, July 9-11, 2018. 2018 Joint Propulsion Conference, AIAA Propulsion and Energy Forum, Cincinnati, OH, July 9-11, 2018. AIAA 2018-4940 paper. https://doi.org/10.2514/6.2018-4940 , Jul-2018
Papers from Meeting Proceedings	Kartuzova O, Kassemi M. "CFD Prediction of Microgravity Self-Pressurization and Pressure Control and Validation against a Tank Pressure Control Experiment." The 53rd AIAA/SAE/ASEE Joint Propulsion, Atlanta, Georgia, July 10–12, 2017. Proceedings of the 53rd AIAA/SAE/ASEE Joint Propulsion, Atlanta, Georgia, July 10–12, 2017. AIAA 2017-4660 paper. https://doi.org/10.2514/6.2017-4660 , Jul-2017
Papers from Meeting Proceedings	Kartuzova O, Kassemi M. "Modeling K-Site LH2 Tank Chilldown and No Vent Fill in Normal Gravity." 53rd AIAA/SAE/ASEE Joint Propulsion, Atlanta, Georgia, July 10–12, 2017. Proceedings of the 53rd AIAA/SAE/ASEE Joint Propulsion, Atlanta, Georgia, July 10–12, 2017. AIAA 2017-4662 paper. https://doi.org/10.2514/6.2017-4662 , Jul-2017
Significant Media Coverage	Kassemi M. "Webinar: Results of the Zero-Boil-Off Tank (ZBOT) ISS Experiment - PART I - Microgravity Self-Pressurization." NASA Engineering and Safety Center (NESC) Academy Webinar, July 2018., Jul-2018